

Geometric Verification & Scene Graph Augmentation

Assignment 1, Problem 2 — Understanding COLMAP's SfM Pipeline

Team [Names]

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Task: validate matched image pairs and build a robust scene graph

- **Paper:** Schönberger & Frahm, *Structure-from-Motion Revisited*, CVPR 2016 – Sec. 2.1 & 4.1
- **Code:** `colmap/src/colmap/estimators/two_view_geometry.{h,cc}`

Five guiding questions we answer, each grounded in code:

- 1 What does H verify vs. F/E, and why doesn't a valid H mean "safe to triangulate"?
- 2 Why isn't a single model (just F) enough?
- 3 What do N_H/N_F and N_E/N_F actually indicate about the scene?
- 4 What is a WTF (watermark) pair, and why does it fool a naive verifier?
- 5 With fixed thresholds, how does a borderline pair get resolved?

Required experiment (two parts): real N_F , N_E , N_H across scene types, plus a borderline-pair threshold sweep.

The Verification Decision Tree (the code, in one slide)

TwoViewGeometryOptions defaults — [two_view_geometry.h:18-111](#):

<code>min_num_inliers</code>	<code>= 15</code>	floor for any non-degenerate model
<code>min_E_F_inlier_ratio</code>	<code>= 0.95</code>	paper's ϵ_{EF}
<code>max_H_inlier_ratio</code>	<code>= 0.80</code>	paper's ϵ_{HF}
<code>watermark_min_inlier_ratio</code>	<code>= 0.70</code>	WTF border+translation ratio
<code>ransac_options.max_error</code>	<code>= 4.0px</code>	shared by F, E, H

Classification, [two_view_geometry.cc:1148-1207](#) (calibrated) / [cc:425-441](#) (uncalibrated):

```
fit E, F, H independently (LO-RANSAC, same 4px thresh.)
if E passed and N_E/N_F > eps_EF:
    if N_H/N_E > eps_HF: config = PLANAR_OR_PANORAMIC
    else: config = CALIBRATED
elif F passed:
    if N_H/N_F > eps_HF: config = PLANAR_OR_PANORAMIC
    else: config = UNCALIBRATED
elif H passed: config = PLANAR_OR_PANORAMIC
else: config = DEGENERATE
```

Q1: What does H verify vs. F/E?

Homography H

- Valid only if the scene is a single plane, *or* the camera underwent pure rotation (zero baseline)
- A 2D projective map – no notion of depth at all

Fundamental / Essential (F/E)

- Valid for *any* 3D scene, as long as the camera translates
- Encodes depth-dependent disparity along epipolar lines

Why a valid H fit $\neq$ safe to triangulate:

- Pure rotation: zero baseline $\Rightarrow$ triangulation is undefined. Code explicitly sets `tri_angle=0` for PANORAMIC and the paper states it does *not* triangulate panoramic pairs (Sec. 4.1).
- Planar: only in-plane structure is constrained; nothing is said about off-plane depth.
- **Found on real data:** our calibrated Middlebury stereo pair (genuine depth variation, confirmed by its ground-truth disparity map) still gets $H_F\text{-ratio} = 0.98$ — a real scene that *looks* planar to the ratio test purely because its baseline is narrow relative to scene depth.

Q2: Why isn't checking F alone enough?

- A pure-planar or pure-rotation data set doesn't just *fail* to fit F — it satisfies **many different F's equally well** (F is poorly constrained / near non-unique when every correspondence also lies exactly on a homography).
- So N_F alone can look completely healthy (plenty of “inliers”, low residual) even though the recovered F is spurious and its epipolar geometry & pose decomposition are meaningless.

The fix, in code ([two_view_geometry.cc:1191-1199](#)):

```
elif RansacReportPassed(F_report, min_num_inliers):  
    num_inliers = F_report.num_inliers  
    if H_F_inlier_ratio > options.max_H_inlier_ratio:  
        config = PLANAR_OR_PANORAMIC # <- override, even though F "passed"  
    else:  
        config = UNCALIBRATED
```

Checking H *alongside* F catches exactly this failure mode: F “passing” its own threshold is overridden the moment H explains almost as many matches.

Q3: What do N_H/N_F and N_E/N_F indicate?

$$N_H/N_F > \epsilon_{HF} = 0.80$$

A 2D projective map explains the pair almost as well as full epipolar geometry $\Rightarrow$ **planar scene or pure rotation**. Later split into PLANAR vs. PANORAMIC by decomposing H and checking if the recovered translation norm ≈ 0 ([cc:985-992](#)).

$$N_E/N_F > \epsilon_{EF} = 0.95 \text{ (calibrated pairs only)}$$

The essential matrix (built from the *given* focal-length prior) explains nearly as many inliers as the fully general fundamental matrix $\Rightarrow$ **the assumed calibration is trustworthy** $\Rightarrow$ use CALIBRATED. If it fails, COLMAP falls back to UNCALIBRATED, discarding the possibly-wrong prior.

We verify both ratios directly on real data in the experiment (next section).

Q4: What is a WTF pair?

A pair whose only reliable matches come from an **identical overlay burned into both images** (timestamp, watermark, logo) at the same screen position — not real scene overlap.

Detection, `DetectWatermarkMatches`, `cc:1550-1615`:

- 1 $\geq 70\%$ of the winning model's inliers lie in the image **border** (outside a 10%-of-diagonal margin, on *both* images)
- 2 That same fraction is explained by a **pure 2D translation** (RANSAC-fit), not a general H or F/E

Why a naive verifier is fooled: a pure translation is a *degenerate special case of both H and F/E* — it passes ordinary geometric verification with a low residual and a healthy inlier count. Only the border-localization + translation-only check reveals it's fake.

Constructed on real data: two unrelated photos, identical stamped corner text $\Rightarrow$ `is_watermark=True`, border ratio = 0.91, translation-inlier ratio = 0.90.

Q5: Borderline pairs & fixed thresholds

- A pair with a large flat surface *mixed* with genuine off-plane clutter can land N_H/N_F right next to $\epsilon_{HF} = 0.80$.
- With a **hard** fixed threshold, this ambiguity is *not* softly resolved: a handful of extra/fewer RANSAC inliers (different random seed, slightly more texture) flips the entire pair between
 - UNCALIBRATED – general, safe to triangulate broadly, or
 - PLANAR_OR_PANORAMIC – triangulation skipped for this pair
- No intermediate/soft handling exists at this stage; downstream robustness (seed-pair scoring, BA outlier filtering) is the only cushion, and it's indirect.

Demonstrated directly with a synthetic two-plane scene and a threshold sweep — see experiment.

Experiment Design

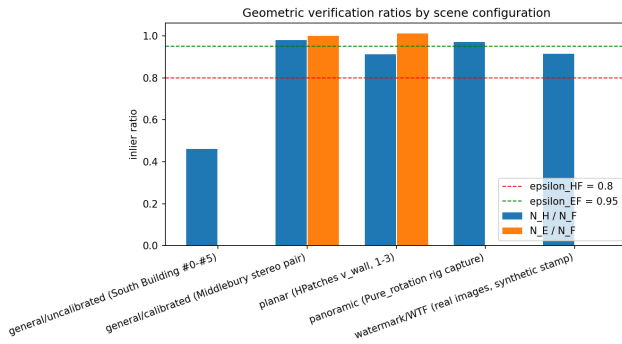
Reimplemented COLMAP's classification (SIFT $\rightarrow$ Lowe ratio test $\rightarrow$ independent LO-RANSAC fits of F/H/(E)) in Python/OpenCV, using COLMAP's *exact* default thresholds, run on four real datasets:

Category	Dataset	Ground truth
General / uncalibrated	COLMAP South Building (128 imgs)	–
General / calibrated	Middlebury-style stereo pair	exact K , disparity map
Planar	HPatches v_wall sequence	exact homographies $H_{1 \rightarrow k}$
Panoramic	Real rig capture, pure rotation	2467-pose GT log
Watermark/WTF	Two real photos + synthetic stamp	constructed

Two required-experiment parts:

- (1) N_F, N_E, N_H across all five categories $\rightarrow$ do the ratios classify them correctly?
- (2) Borderline pair, sweep ϵ_{HF} itself $\rightarrow$ show the exact flip

Experiment 1: Ratios Across Scene Configurations



Pair	N_F	N_E	N_H	N_H / N_F	Config
South Building (general)	26	—	12	0.46	UNCALIBRATED
Middlebury (calibrated)	1842	1845	1807	0.98	PLANAR_OR_PANORAMIC
HPatches v_wall (planar)	492	498	449	0.91	PLANAR_OR_PANORAMIC
Pure-rotation rig (panoramic)	139	—	135	0.97	PLANAR_OR_PANORAMIC
Watermark (real + stamp)	58	—	53	0.91	PLANAR_OR_PANORAMIC, WTF

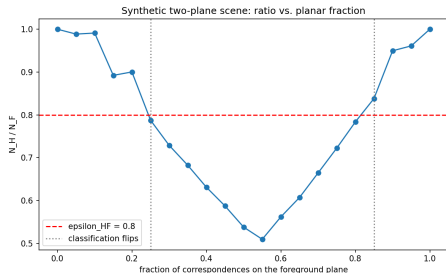
Only the wide-baseline South Building pair falls below $\epsilon_{HF} = 0.80$.

Real-Data Highlight: A Genuine False Positive

- Middlebury pair has **real depth variation** — proven by its shipped ground-truth disparity map (`disp0GT.pfm`)
- Yet: $H_F\text{-ratio} = 0.98$, $H_E\text{-ratio} = 0.98 \Rightarrow \text{PLANAR_OR_PANORAMIC}$
- **Why:** stereo baseline (59.6 mm) is narrow relative to scene depth $\Rightarrow$ a homography fits almost as well as the true epipolar geometry at a 4px threshold

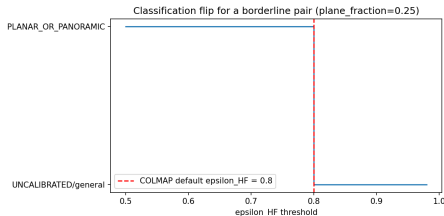
This is a real, data-backed instance of Q1's answer: **a valid H fit is a statistical symptom, not proof of true planarity** – and it directly explains why COLMAP treats N_H/N_F as a heuristic, not a certainty.

Experiment 2: Borderline Pair & Threshold Flip



Synthetic two-plane scene: continuous control of N_H/N_F via the fraction of points on one plane. Two clean flips bracket $\epsilon_{HF} = 0.80$.

Directly answers Q5: a borderline pair's classification is entirely at the mercy of where the fixed threshold happens to sit – there is no soft margin.



Fix the pair whose ratio (0.79) sits right next to the real default; sweep ϵ_{HF} itself $\rightarrow$ flips exactly between 0.78 and 0.80.

Bonus: When Even Self-Calibration Fails

Tried to recover K for the panoramic pair from real ground-truth rotations ($H = KRK^{-1}$, Hartley & Zisserman Ch. 8) – it's provably impossible here.

- Every relative rotation in the rig capture has axis $\approx [0, 0, \pm 1]$ in the camera's own frame: a **pure roll about the optical axis**, not a pan/tilt
- For a roll about the optical axis: $K R_z(\theta) K^{-1} = R_z(\theta)$ **exactly, for every focal length** – verified numerically to 10^{-14}
- Self-calibration fit confirms it: fitted focal swings from ~ 50 px to ~ 8500 px depending on which frame pairs are used, hugging search bounds

Punchline: a second, independent, data-backed reason (beyond zero baseline) the paper never triangulates from panoramic pairs – this sequence can't even be *calibrated* from, let alone reconstructed.

Takeaways

- Multi-model verification (H *and* F *and* E) exists because a **single model can't tell a genuine general scene from a degenerate one** – F alone can be well-fit even when it's meaningless.
- The ratios N_H/N_F , N_E/N_F are heuristics against fixed thresholds, not proofs – confirmed by a real narrow-baseline stereo pair being misclassified as planar/panoramic despite genuine depth variation.
- WTF detection needs a *third*, spatially-localized check (border + pure translation) because a translation is a blind spot shared by both H and F/E.
- Borderline pairs flip discretely at the threshold – no soft margin exists in this stage of the pipeline.
- Panoramic (pure-rotation) pairs are excluded from triangulation for two independent reasons: zero baseline, *and* (shown here) the sequence itself may not even be self-calibratable.

Backup: Code Reference Index

All paths relative to `colmap/src/colmap/`.

What		Where
Default thresholds (TwoViewGeometryOptions)		<code>.../two_view_geometry.h:18-111</code>
Calibrated classification (E, F, H)		<code>.../two_view_geometry.cc:1075-1235</code>
Uncalibrated classification (F, H)		<code>.../two_view_geometry.cc:377-466</code>
Watermark / WTF detection		<code>.../two_view_geometry.cc:1550-1615</code>
Planar vs. panoramic split		<code>.../two_view_geometry.cc:985-992</code>
ConfigurationType enum		<code>scene/two_view_geometry.h:16-41</code>

Paper: Schönberger & Frahm, *Structure-from-Motion Revisited*, CVPR 2016, Sec. 2.1 & 4.1.

Code & experiment: `problem2_geometric_verification/` – `two_view_geometry.py`,
`real_datasets.py`, `run_experiment.py`, `pure_rotation_self_calibration.py`.